# llmXive Automated Review of OPID: On-Policy Skill Distillation for Agentic Reinforcement Learning

*llmXive automated review panel*

**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a novel method (OPID) for on-policy skill distillation, with claims supported by internal numerical consistency (e.g., Table 1 gains match the text descriptions). However, a critical issue arises with the citation of baselines. The paper targets ICLR 2025 (indicated by the bibliography style) but cites multiple key baselines (OPSD, Skill-SD, RLSD, SDAR) as 2026 publications (e.g., zhao2026opsd, wang2026skillsd). These future-dated references cannot be verified as existing works for a 2025 submission, rendering the comparative claims against these baselines unsupported. The authors must either correct the publication years to match existing literature or replace these baselines with verifiable, contemporaneous works. Additionally, the claim regarding "Qwen3-1.7B" relies on a model (yang2025qwen3) that may also be future-dated relative to the paper's timeline, requiring verification of the model's actual availability.

**Action items.**

- [**science**] The bibliography uses 'iclr2025_conference' style, implying a 2025 submission, yet cites multiple baselines (e.g., OPSD, Skill-SD, RLSD, SDAR) with 2026 publication years (e.g., zhao2026opsd, wang2026skillsd). These future-dated references cannot be verified as existing baselines for a 2025 paper, invalidating the comparative claims. Verify the actual publication years or replace with existing baselines.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_figure_critic.md`

### Figure 1

The figure presents a clear bar chart comparing methods across three tasks, but the caption's claim that OPID is 'competitive' on Search-based QA is misleading given that several baselines outperform it. Additionally, the method names on the y-axis lack explicit definition in the caption or legend.

### Figure 2

Figure 2 provides a clear and comprehensive visual overview of the OPID framework, effectively illustrating the three-stage process from skill extraction to routing and self-distillation. The diagram is well-structured with distinct color coding and clear labels that align perfectly with the provided caption, making the complex methodology easy to follow.

## Figure 3

Figure 3 effectively visualizes the training dynamics of OPID and GRPO with clear axes, units, and a legend. The distinction between raw measurements (translucent) and smoothed trends (solid) is explicitly defined in the caption and visually distinct, supporting the claim of improved performance.

## Figure 4

The figure effectively displays the performance gap between OPID and GRPO across data fractions, but the caption's claim about approaching full-data performance at 60% is not strongly supported by the visual gap shown, and the ' 40% Data' annotation lacks clear context.

## Figure 5

Figure 5 effectively illustrates the qualitative differences between GRPO and OPID agents on the specified ALFWorld task. The side-by-side trajectory comparison is clear, with distinct annotations highlighting specific failure modes (hallucination, substitution) and successful steps, fully supporting the claims in the caption.

## Figure 6

Figure 6 presents a time-series of critical step counts but mislabels the y-axis as an average when the caption describes per-trajectory counts, and the annotation lacks contextual definition.

## Figure 7

Figure 7 presents two line plots comparing advantage signals, but the x-axis unit is undefined and the y-axis scaling in the bottom panel is ambiguous, hindering accurate interpretation of the training dynamics.

## Figure 8

Figure 8 displays the 'Analyser Prompt' used in the study. The text is legible, and the figure effectively communicates the instructions and JSON schema required for the analyzer component described in the paper.

## Figure 9

Figure 9 presents a clear, step-by-step text trajectory of an agent completing a task in the ALFWorld environment. The layout is readable, the steps are logically ordered, and the content aligns perfectly with the caption's description of a full trajectory.

### Figure 10

Figure 10 presents a clear, text-based qualitative example of an agent trajectory on ALFWorld, consistent with the format of Figure 9 and the caption description. The step-by-step breakdown of observations, reasoning, and actions is legible and effectively illustrates the agent's behavior.

### Figure 11

Figure 11 presents a clear, text-based visualization of a Search-QA agent trajectory, effectively illustrating the step-by-step reasoning and search actions described in the caption. The layout is uncluttered, and the content is fully legible, successfully supporting the claim of showing a full trajectory.

### Figure 12

Figure 12 presents a clear, step-by-step textual log of an agent's trajectory on a Search-QA task. The layout is readable, the reasoning steps are distinct, and the final answer is explicitly marked, fully supporting the caption's description.

**Action items.**

- [**science**] Figure 1: The caption claims OPID is 'competitive' on Search-based QA, but the chart shows OPID (49.2) is significantly outperformed by multiple baselines (e.g., Skill-GRPO* at 47.5, Skill-SD at 47.8, RLSD at 49.0, SDAR at 49.0) and is not the strongest; the claim contradicts the visual data.
- [**writing**] Figure 1: The y-axis labels (method names) are not explicitly defined in the caption or legend, forcing the reader to infer which bars correspond to which method groups without a clear key.
- [**science**] Figure 4: The caption claims OPID 'approaches full-data GRPO performance using about 60% of the data,' but the chart shows OPID at 60% data (~73%) is still significantly higher than GRPO at 100% data (~76%), and the red arrow annotation points to the gap between OPID at 60% and GRPO at 100% rather than showing convergence. The visual evidence does not support the specific claim of 'approaching' performance at that data fraction.
- [**writing**] Figure 4: The annotation ' 40% Data' with a downward arrow is ambiguous; it is unclear if this refers to the data reduction (100% - 60% = 40%) or a specific metric value, and the arrow placement between the curves does not clearly link to the x-axis value of 60%.
- [**science**] Figure 6: The y-axis label 'Avg critical steps' contradicts the caption's claim that the curve reports 'how many timesteps are selected... in each trajectory' (a count per sequence). The y-axis values (~3.0–4.0) are too low to represent a raw count of steps per trajectory if the x-axis is 'Training step' (1–150), suggesting the metric is an average over multiple trajectories per step, but this aggregation is not explained in the caption or axis label.
- [**writing**] Figure 6: The annotation 'avg 3.692' is placed near the end of the plot without

clarifying whether it represents the global mean across all training steps or a local average; the caption does not define this value.

- [**science**] Figure 7: The x-axis is labeled 'Training step' with a range of 1–150, but the caption describes 'magnitudes... during OPID training' without specifying the unit of the step (e.g., iterations, epochs, or gradient updates). This ambiguity makes it impossible to assess the training duration or convergence rate.
- [**writing**] Figure 7: The y-axis label for the bottom panel, 'Skill abs advantage(1e-4)', uses scientific notation that implies the plotted values are scaled by 10,000. However, the axis ticks (1.5–4.0) and the caption's description of 'magnitudes' suggest these are raw values. If the values are indeed scaled, the caption should explicitly state this to avoid misinterpretation of the signal's strength relative to the episode advantage.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally accessible to a competent reader from an adjacent field (e.g., standard NLP or RL), as core concepts like "on-policy," "distillation," and "advantage" are used in their standard disciplinary sense. However, there are specific instances of undefined notation and shorthand that would stall a reader not deeply embedded in this specific subfield's recent literature.

First, in Section 3.1, the routing masks are defined using the symbol `ind` (e.g., $\ind[t\in\mathcal{C}_\tau]$). While mathematically standard for an indicator function, it is not defined in the text or the notation appendix. A reader from a different background might momentarily pause to confirm if this is a specific operator or a typo.

Second, in Section 3.3, the skill-based advantage equation includes a term $m_{\tau,t,\ell}$. This variable is never defined in the surrounding text. It is likely a mask for valid tokens (excluding padding or special tokens), but without an explicit definition, the equation is technically incomplete for an outsider.

Third, the term "episode-relative advantage" in Section 3.4 is slightly ambiguous. The formula $(R(\tau)-\mu_q)/\sigma_q$ describes a group-relative normalization (common in GRPO), but the label "episode-relative" could be confused with standard return normalization over a single episode's steps. A brief clarification of the normalization scope would prevent misinterpretation.

Finally, the "analyzer" $\mathcal{A}$ is introduced as a black box. While the context implies it is an LLM, explicitly stating "a separate, frozen LLM-based analyzer" at first use would clarify the architectural distinction between the training policy and the distillation signal generator.

These are minor, easily fixable omissions that do not reflect a lack of rigor but rather a failure to explicitly define symbols that are obvious to the authors.

**Action items.**

- **[writing]** Section 3.1 (Problem Formulation): The symbol `ind` is used in the definition of routing masks ($q^{\mathrm{step}}_{ au,t}=\ind[t\in\mathcal{C}_ au]$) without definition. Define it as the indicator function (1 if true, 0 otherwise) at first use.
- **[writing]** Section 3.3 (Critical-First Skill-Conditioned Self-Distillation): The variable `m` in the advantage equation ($A^{\mathrm{skill}} = \dots m_{ au,t,ll}$) is undefined. Specify that $m_{ au,t,ll}$ is a token mask (e.g., excluding padding or non-generation tokens).
- **[writing]** Section 3.4 (Policy Optimization): The term `episode-relative advantage` is used for $A^{\mathrm{ep}}_{ au}$, but the equation shows it is computed per trajectory within a group. Clarify if this is a 'group-relative advantage' or define the 'episode' scope explicitly to avoid confusion with standard episode-level returns.
- **[writing]** Section 3.2 & 3.3: The term 'analyzer' ($\mathcal{A}$) is introduced as an LLM-based component. While the function is clear, explicitly state in the first mention that $\mathcal{A}$ is a separate, frozen LLM instance used for offline trajectory analysis to distinguish it from the policy $\pi_ heta$.

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is largely sound, with the proposed method (OPID) logically derived from the identified problem (sparse rewards in agentic RL) and the solution (on-policy skill distillation). The definitions of episode-level and step-level skills are consistent throughout the Methods and Appendix sections. The mathematical formulation of the advantage function (combining outcome and skill advantages) follows directly from the preceding definitions.

However, there are specific instances where the numerical claims in the text do not entail from the data presented in the tables, representing a break in the internal consistency of the argument:

1. **Numerical Mismatch in Main Results:** In Section 4.2 ("Main Results"), the text states: "On Qwen3-1.7B: +12.8 (ALFWorld), +26.5 (WebShop)."

- **Premise (Table 1):** For Qwen3-1.7B on WebShop, the table lists OPID Score as **85.0** and GRPO Score as **67.3**.
- **Calculated Difference:** $85.0 - 67.3 = 17.7$.
- **Conclusion (Text):** The text claims a gain of **+26.5**.
- **Gap:** The conclusion (+26.5) does not follow from the premise (Table 1 values). This suggests either a calculation error in the text, a typo in the table, or a mismatch in the data used for

the text summary versus the table. This is a clear non-entailment.

1. **Generalization vs. Specific Data:** The text in Section 4.2 asserts “OPID improves over GRPO in most combinations” and lists specific positive gains. While “most” allows for exceptions, the paragraph immediately follows with specific numbers that are presented as definitive improvements. The presence of a clear degradation in the “Pick” task for Qwen3-1.7B (65.9 vs 71.1) in the very same table creates a minor logical friction. While not a fatal contradiction, the text could be more precise (e.g., “improves in the majority of cases, with notable exceptions in...”) to align the generalization with the specific data points shown.
2. **Ablation Consistency:** The ablation studies (Section 4.4) and their corresponding tables (Table 3 and Table 4) are internally consistent. The text correctly interprets the drops in performance when removing components, and the numbers in the text match the tables.

The primary issue is the numerical discrepancy in Section 4.2. Correcting the text to match the table (or vice versa) is required to restore logical consistency between the evidence presented and the claims made.

**Action items.**

- [**writing**] Section 4.2 (Main Results) claims OPID improves over GRPO by ‘+26.5 (WebShop)’ on Qwen3-1.7B. However, Table 1 shows OPID (85.0) vs GRPO (67.3), a difference of +17.7. The text’s number (+26.5) does not follow from the table’s data. Verify the calculation or correct the text to match the table.
- [**writing**] Section 4.2 states ‘OPID improves over GRPO in most combinations’ and cites specific gains. However, Table 1 shows that on Qwen3-1.7B for the ‘Pick’ task in ALFWorld, OPID (65.9) is lower than GRPO (71.1). The text’s generalization ‘improves... in most combinations’ is technically true but the specific claim of consistent improvement in the paragraph context ignores this clear counter-example in the same table, creating a slight logical tension between the summary and the data presentation.

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims regarding the scope of its results that slightly exceed the demonstrated evidence, primarily in the areas of generalization and consistency.

First, the **Abstract** and **Introduction** assert that OPID improves “robustness” and “generalizes” to broader settings. The only evidence provided for generalization is the “ALFWorld Unseen” split (Section 3.4, Figure 4). While this is a valid test of distribution shift within a single environment, the paper does not demonstrate cross-domain generalization (e.g., training on ALFWorld and testing on WebShop) or robustness to entirely different task families. The claim of generalization should be scoped to "unseen splits within the same environment" or supported by cross-domain experiments.

Second, the **Conclusion** and **Main Results** (Section 3.2) describe the improvements as "consistent." While the *average* performance improves, Table 1 reveals specific instances where OPID underperforms the GRPO baseline. For example, on the Qwen2.5-3B model, OPID scores 88.9 on ALFWorld "Clean" compared to GRPO's 96.2. Similarly, on Qwen3-1.7B, OPID scores lower than GRPO on NQ (38.1 vs 40.0) and TriviaQA (58.1 vs 58.9). The narrative frames the method as uniformly superior, omitting these specific failure modes. A more accurate claim would acknowledge that improvements are "generally observed" or "on average," while noting that performance varies by task type and model size.

Finally, the **Qualitative Analysis** (Section 3.5) presents a single case study (Figure 3) where OPID succeeds and GRPO fails. While illustrative, presenting this as the primary evidence for "reduced invalid behaviors" without quantifying the frequency of such failures across the test set risks cherry-picking. The text should clarify that this is an *illustrative* example of the mechanism, not a statistical proof of reduced hallucination rates, unless such a metric is explicitly reported in the tables.

These are primarily rhetorical overreaches that can be fixed by tightening the language in the abstract and conclusion to match the specific boundaries of the experimental data.

**Action items.**

- [**writing**] Abstract claims 'generalizes' broadly, but evidence is only ALFWorld 'unseen' split (Sec 3.4). Scope claim to 'unseen splits within same environment' or add cross-domain tests.
- [**writing**] Conclusion states 'consistent improvements' but Table 1 shows OPID loses on specific tasks (e.g., ALFWorld Clean on 3B, NQ on 1.7B). Qualify 'consistent' to 'generally' and acknowledge variance.
- [**writing**] Qualitative analysis (Fig 3) presents one success case as evidence for 'reduced invalid behaviors' without statistical backing. Clarify this is illustrative, not a quantitative proof of reduced hallucination.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a reinforcement learning method (OPID) for agentic LLMs using on-policy skill distillation. From a safety and ethics perspective, the work is low-risk. The research utilizes standard, public benchmarks (ALFWorld, WebShop, Search-based QA) and does not involve human subjects, personal data, or sensitive information. The methodology focuses on improving agent efficiency and sample complexity in simulated environments, with no dual-use capabilities identified that would meaningfully lower the barrier to harmful activities (e.g., automated cyberattacks, disinformation generation, or biological synthesis).

The paper does not release any new datasets containing PII or scraped content that would violate terms of service; the training data consists of interactions within the specified benchmark environments. The "analyzer" component described in the method uses an LLM to extract skills from trajectories, but this is an internal training mechanism, not a system designed for deception, surveillance, or manipulation of human users. There are no undisclosed conflicts of interest or operational vulnerabilities disclosed that require responsible disclosure.

As this is a third-party preprint, the absence of a formal "Broader Impacts" statement is noted but does not constitute a safety failure given the benign nature of the research (algorithmic improvement on standard benchmarks). The paper does not require specific safety mitigations or disclosures beyond what is standard for this subfield.

## Scientific Evidence

*Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling method (OPID) for enhancing agentic RL via on-policy skill distillation. However, the evidentiary strength of the central claims is currently undermined by a lack of statistical rigor in the experimental design.

The primary concern is the absence of variance reporting. Table 1 presents headline performance numbers (e.g., 84.3% vs 75.0% on ALFWorld) derived from what appears to be single training runs. In reinforcement learning, performance is highly sensitive to random seeds, hyperparameter initialization, and rollout stochasticity. A single-point improvement of ~9% is well within the range of variance observed in standard RL baselines. Without reporting results across multiple seeds (e.g., 3-5) with mean and standard deviation, the reader cannot determine if the reported gains are a robust effect of the method or a lucky seed. This is a critical gap for a paper claiming "consistent improvements."

Secondly, the ablation studies do not fully isolate the proposed mechanisms. The "w/o Routing" ablation (Table 4) removes the critical-first logic but leaves the skill extraction and injection pipeline intact. It is unclear if the performance drop is due to the loss of the routing logic or simply the loss of the specific set of "critical" steps identified by the analyzer. A more rigorous control would inject step-level skills uniformly or randomly to distinguish the value of *routing* from the value of *having* step-level skills.

Finally, the reliance on an external LLM analyzer (GLM-5.2) introduces a potential confound. The paper attributes success to "on-policy" skills, but the analyzer is a separate, potentially stronger model. It is possible the improvement stems from the analyzer's external knowledge or superior reasoning rather than the distillation of the policy's own trajectory. A negative control using a non-intelligent or random text generator as the analyzer would help confirm that the *content* of the extracted skills is the driver, not just the presence of an auxiliary context.

Addressing these design gaps—specifically by adding seed variation and tighter ablation controls—is necessary to substantiate the claim that OPID provides a stable, generalizable improvement over outcome-only RL.

**Action items.**

- **[writing]** The paper presents a compelling method (OPID) for enhancing agentic RL via on-policy skill distillation. However, the evidentiary strength of the central claims is currently undermined by a lack of statistical rigor in the experimental design. The primary concern is the absence of variance reporting. Table 1 presents headline performance numbers (e.g., 84.3% vs 75.0% on ALFWorld) derived from what appears to be single training runs. In reinforcement learning, performance is highly sensitive to r

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment of results in this paper is insufficient to support the inferential claims made. While the experimental design appears sound, the **reporting of quantitative results** lacks necessary uncertainty measures.

**1. Missing Uncertainty Reporting:** Throughout Section 4 and Tables 1, 3, and 4, performance metrics are reported as single point estimates (e.g., "92.7%"). There is no mention of the number of random seeds, nor are standard deviations (SD), standard errors (SE), or confidence intervals (CI) provided. In RL/LLM training, performance is highly sensitive to initialization. Reporting a single number creates false precision and makes it impossible to determine if differences (e.g., +9.3 points) are significant or due to random variance. The field standard is `mean ± SD` over at least 3-5 independent runs.

**2. Unsubstantiated Claims of Significance:** The text uses language implying statistical significance (e.g., "consistently strengthens," "improves") based solely on point estimate differences. No hypothesis tests (paired t-test, Wilcoxon, bootstrap) are described or reported. Without p-values or CIs, these claims are descriptive observations, not statistical inferences. Authors must either perform tests on seed-level data and report results or soften language to reflect observed improvements without statistical validation.

**3. Multiple Comparisons in Ablations:** Ablation studies (Tables 3 and 4) involve multiple pairwise comparisons. The lack of variance reporting makes it difficult to assess the reliability of observed drops. If authors claim a component is "essential" based on a performance drop, they must demonstrate robustness across seeds.

**Recommendation:** Re-run experiments with multiple random seeds (min 3, preferably 5) and report mean and SD for all main results and ablations. Apply appropriate statistical tests to seed-level data to validate improvement claims. If re-running is not feasible, remove all language implying statistical significance and state results are from single runs or unverified means.

**Action items.**

- [**science**] Table 1 and Section 4.2 report single-point accuracy/success rates without variance (SD/SE/CI) or seed counts. RL results are stochastic; reporting single runs implies false precision. Report mean ± SD over  3 seeds for all main results to assess stability.
- [**writing**] Section 4.2 claims OPID 'improves' or is 'better' based on point estimates alone, with no hypothesis test (t-test, Wilcoxon, bootstrap) or p-values reported. Without statistical validation, these are descriptive, not inferential. Run tests on seed data or rephrase to 'observed improvement'.
- [**writing**] Ablation Tables 3 and 4 compare OPID to variants using single-point averages. With multiple comparisons, false positives are possible. Report variance across seeds or acknowledge multiplicity to ensure observed drops (e.g., -10.2 points) are robust and not noise.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured, with a logical flow from problem formulation to method, results, and analysis. The abstract effectively sets the stage, though it could be more specific about the quantitative gains. The introduction clearly delineates the gap in current methods and the proposed solution.

However, several sections suffer from minor clarity issues that require a second read. In Section 3.1, the transition from the standard RL objective to the group-relative advantage used in GRPO/OPID is abrupt; the reader must infer the connection between the expectation over trajectories and the group-based baseline calculation. Similarly, in Section 3.3, the introduction of the token index   in the log-probability equations appears without prior definition in the immediate context, forcing the reader to backtrack to understand the scope of the advantage calculation.

The "Main Results" section (4.2) includes a paragraph discussing the "internalization" of skills that references a baseline ("Skill-GRPO") without a clear, immediate definition of what that baseline entails in this specific context, slightly obscuring the comparison. Additionally, the notation in the Appendix (A.1) uses q_i for the teacher, while the main text uses q^step/q^ep; a brief cross-reference would prevent confusion for readers navigating between the theoretical appendix and the method section.

Finally, the abstract's concluding sentence lists the benchmarks but fails to mention the specific performance gains or the unique "hindsight" mechanism that drives the results, missing an opportunity to summarize the paper's primary contribution more concretely. Addressing these specific points will significantly improve the readability and flow of the manuscript.

**Action items.**

- [**writing**] The paper is generally well-structured, with a logical flow from problem formulation to method, results, and analysis. The abstract effectively sets the stage, though it could be more specific about the quantitative gains. The introduction clearly delineates the gap in current methods and the proposed solution. However, several sections suffer from minor clarity issues that require a second read. In Section 3.1, the transition from the standard RL objective to the group-relative advantage used i